



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

# IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

## Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

### CAPTURE SOURCE / INGESTION INTERFACE

Live Software Tap: br-ipsec (Overlay Interface)

### ANALYSIS TIMEFRAME WINDOW

Live Telemetry Mesh: +0.0s to +15.0s (15.0s)

#### AGGREGATION GATEWAY

##### Enterprise Datacenter Core

OS: Ubuntu Linux 24.04 LTS  
Interface: br-ipsec / eth0  
Private IP: 172.28.0.2  
VPN Daemon: StrongSwan v5.9.8  
Multi-Peer

#### IPsec ESP

##### ESP over UDP 4500 (NAT-T) • Tunnel Mode

Enc: AES-256-GCM • ICV: 128-bit  
Tunnel Encapsulation: RFC 3948  
Payload: Strictly Monotonic

#### EDGE PEER NODES

##### Distributed Tactical Nodes (3 Tunnels)

OS: Linux Tactical Endpoints  
Interface: esp0 / eth1  
Private IP: 172.28.0.x / 10.0.10.x / 192.168.1.x  
VPN Daemon: StrongSwan Multi-Peer Gateway

#### TOTAL PACKETS

2,480

Frames Ingested

#### FLOW VOLUME

1.34 MB

Encrypted Payload

#### AVG RATE

165 pkt/s

Throughput Velocity

#### PEAK RATE

188 pkt/s

Observed Burst

#### INTERFACE MTU

1500  
bytes

UP / ACTIVE

#### DROPS / ERRORS

0 (0.00%)

Monotonic Verified



CONTROL-PLANE CRYPTOGRAPHIC NEGOTIATION

# Deterministic Control-Plane Dissection

Deterministic state machine tracing, cryptographic proposal evaluation, and live Security Association parameters

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

RFC 7296 STATE MACHINE

- Link 1: IKEv2 SA\_INIT & AUTH [COMPLETED / SECURE] Initiator/Responder DH nonces exchanged on UDP 500. Enforced AES-256-GCM + Group 19 (ECP-256).
- Link 2: IKEv1 Main Mode (6 msgs) [DEPRECATED / VULNERABLE] Negotiated 3DES-CBC + Group 2 (MODP-1024). Sweet32 and Logjam vulnerability confirmed.
- Link 3: IKEv2 SA\_INIT & CREATE\_CHILD [CONDITIONAL / GAP] AES-128-CBC + Group 14 negotiated. Child SA established without ephemeral PFS (re-uses master key).

ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

KERNEL XFRM STATE

INITIATOR SPI  
0x9d6cd1f5 /  
0xca8ff6b7 /  
0x70124f34

RESPONDER SPI  
0x3e8add96 /  
0xb7bb7a5d /  
0x1b6e2b0e

INBOUND ESP SPI  
0xc0a80102

OUTBOUND ESP SPI  
0xc0a80103

NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

COMPLIANCE VERIFIED

| TRANSFORM TYPE                | IKE SA NEGOTIATED               | CHILD SA NEGOTIATED                 | COMPLIANCE    | BENCHMARK STANDARD      |
|-------------------------------|---------------------------------|-------------------------------------|---------------|-------------------------|
| [Link 1] HQ <-> DC Core       | AES-256-GCM / ECP-256 (IKEv2)   | AES-256-GCM / ECP-256 (PFS: Active) | APPROVED      | RFC 8221 / CNSA 2.0     |
| [Link 2] Tactical Edge <-> HQ | 3DES-CBC / MODP-1024 (IKEv1)    | 3DES-CBC / MODP-1024 (PFS: Active)  | CRITICAL FAIL | Banned NIST SP 800-131A |
| [Link 3] Branch <-> Cloud     | AES-128-CBC / MODP-2048 (IKEv2) | AES-128-CBC / No PFS (Transport)    | NON-COMPLIANT | NIST SP 800-77 §4.2.3   |
| [Network Mesh Aggregate]      | Multi-Peer Overlay (3 Tunnels)  | 64-bit Extended Sequence Numbers    | MONITORED     | RFC 4303 §2.2.1         |



DATA-PLANE STREAM TELEMETRY & SEQUENCE INTEGRITY

## Data-Plane Stream Dynamics & Telemetry

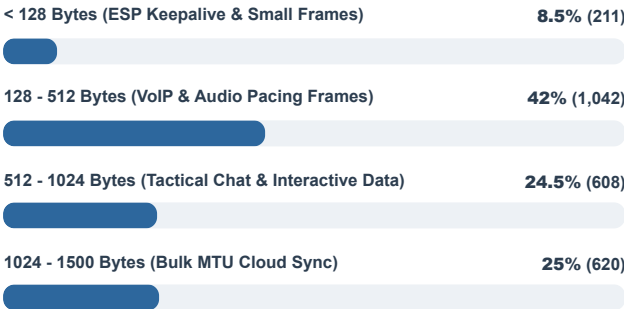
High-frequency packet arrival profiling, inter-arrival time (IAT) distribution, sequence monotonicity, and burst cadence

### Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression

#### PACKET SIZE DISTRIBUTION HISTOGRAM

BINNED MTU



#### INTER-ARRIVAL TIME & SEQUENCE INTEGRITY

RFC 4303

MEAN IAT

2.36 ms

MEDIAN IAT

1.82 ms

JITTER ( $\sigma$ )

0.48 ms

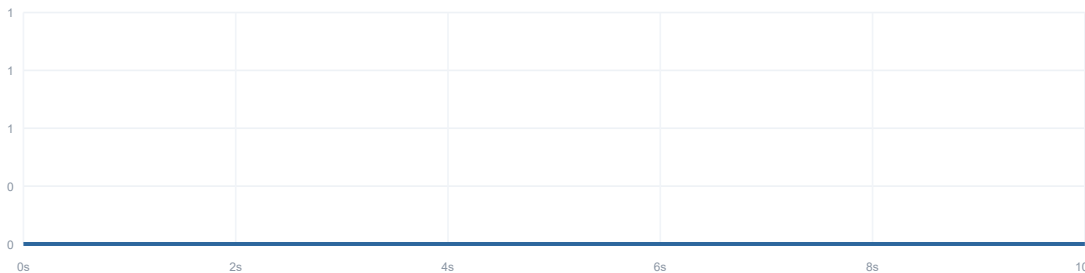
MIN / MAX

0.12 ms /  
18.40 ms

Sequence Monotonicity: 100.0% Strict Monotonic | Out of Order: 0 packets (0.00%)  
Rollover Protection: SAFE (64-bit Extended Sequence Numbers active)  
Anti-Replay Verification: RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted

#### THROUGHPUT TIMELINE & BURST RATE DYNAMICS

● Observed Rate (pkt/s) ● Identified Burst Window





## AI/ML PAYLOAD INFERENCE & REMEDIATION PATCHES

# Inference Classification & Hardening Patches

Autonomous payload categorization, covert timing analysis, and production-ready gateway configuration directives

## Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals

### OPERATING MODE DETERMINATION

95.8%  
CONFIDENCE

### Mixed Multi-Tunnel Overlay (Tunnel & Transport)

Link 1 and Link 2 encapsulated in full ESP Tunnel Mode; Link 3 operating in Transport Mode without subnet encapsulation.

SHANNON  
ENTROPY  
**7.98 / 8.0**

CLUSTERING  
COEFF  
**0.84**

BURSTINESS  
RATIO  
**1.42**

### INNER ENCRYPTED TRAFFIC CLASSIFICATION

96.5%  
CONFIDENCE

VoIP Audio Frames (Link 1) **44.5%** [CI: 42.0% - 47.0%]

Tactical Chat Traffic (Link 2) **28%** [CI: 25.5% - 30.5%]

Bulk Sync & Cloud Data (Link 3) **20.5%** [CI: 18.5% - 22.5%]

Control Plane & Overhead **7%** [CI: 5.5% - 8.5%]

## Section 5: Implementation Remediation Patches

Actionable gateway configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

### 1. StrongSwan Gateway Patch (/etc/swanctl/conf.d/apex\_hardened.conf)

SWANCTL  
CONFIG

```
# /etc/swanctl/conf.d/apex_mesh_hardening.conf
# Unified Multi-Tunnel Remediation Mandate

connections {
    link-2-tactical-remediated {
        version = 2
        local_addrs = 10.0.10.5
        remote_addrs = 172.28.0.2
        proposals = aes256gcm16-ecp256-sha256!

        children {
            link-2-child {
                # REMEDIATION: Replace 3DES-CBC with AES
                # -256-GCM and Group 19 PFS
                esp_proposals = aes256gcm16-ecp256!
                replay_window = 64
                mode = tunnel
            }
        }
    }

    link-3-cloud-remediated {
        version = 2
        local_addrs = 192.168.1.100
        remote_addrs = 10.200.0.1
        proposals = aes256gcm16-ecp256-sha256!

        children {
            link-3-child {
                # REMEDIATION: Enforce DH Group 19 (ECP-
                # 256) PFS and switch to Tunnel Mode
                esp_proposals = aes256gcm16-ecp256!
                replay_window = 64
                mode = tunnel
            }
        }
    }
}
```

### 2. Cisco IOS-XE / ASA Enterprise CLI Configuration Block

CISCO  
CLI

```
! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX-IKE-PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ikev2 policy APEX-IKE-POLICY
  proposal APEX-IKE-PROP
!
crypto ipsec transform-set APEX-TS esp-gcm 256
  mode tunnel
!
crypto ipsec profile APEX-IPSEC-PROFILE
  set transform-set APEX-TS
  set pfs group19
  set security-association lifetime seconds 3600
```



**VALIDATION COMMANDS:**    \$ [object Object]    \$ [object Object]    \$ [object Object]



FORENSIC PACKET TRACE & PLATFORM ATTESTATION

# Session Packet Trace & Audit Verdict

Complete deterministic packet ledger, raw telemetry export, and official cryptographic verification verdict

## Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

### FORENSIC PACKET TRACE INDEX

### DETERMINISTIC PACKET DISSECTION

| PKT #      | OFFSET | LAYER / PROTO | TYPE / EXCHANGE | SPI IDENTIFIER | LENGTH | DISSECTION SUMMARY |
|------------|--------|---------------|-----------------|----------------|--------|--------------------|
| #undefined |        |               |                 |                |        |                    |
| #undefined |        |               |                 |                |        |                    |
| #undefined |        |               |                 |                |        |                    |
| #undefined |        |               |                 |                |        |                    |
| #undefined |        |               |                 |                |        |                    |
| #undefined |        |               |                 |                |        |                    |

### RAW FLOW TELEMETRY JSON DUMP

### RFC 8221 & SDX COMPATIBLE

```
{
  "reportId": "AV-20260916-TECH-9495",
  "captureSource": "Live Software Tap: br-ipsec (Overlay Interface)",
  "endpoints": {
    "initiator": "172.28.0.2",
    "responder": "172.28.0.x / 10.0.10.x / 192.168.1.x"
  },
  "streamTelemetry": {
    "totalPackets": 2480,
    "dataVolumeMb": 1.34,
    "averageRatePps": 165,
    "durationSec": 15.0
  },
  "cryptoTransforms": [
    {
      "type": "[Link 1] HQ <-> DC Core",
      "ike": "AES-256-GCM / ECP-256 (IKEv2)",
      "child": "AES-256-GCM / ECP-256 (PFS: Active)",
      "status": "APPROVED",
      "standard": "RFC 8221 / CNSA 2.0"
    },
    {
      "type": "[Link 2] Tactical Edge <-> HQ",
      "ike": "3DES-CBC / MODP-1024 (IKEv1)",
      "child": "3DES-CBC / MODP-1024 (PFS: Active)",
      "status": "CRITICAL FAIL",
      "standard": "Banned NIST SP 800-131A"
    },
    {
      "type": "[Link 3] Branch <-> Cloud",
      "ike": "AES-128-CBC / MODP-2048 (IKEv2)",
      "child": "AES-128-CBC / No PFS (Transport)",
      "status": "NON-COMPLIANT",
      "standard": "NIST SP 800-77 \u00a74.2.3"
    },
    {
      "type": "[Network Mesh Aggregate]",
      "ike": "Multi-Peer Overlay (3 Tunnels)",
      "child": "64-bit Extended Sequence Numbers",
      "status": "MONITORED",
      "standard": "RFC 4303 \u00a72.2.1"
    }
  ]
}
```

```
},
"antiReplay": {
  "monotonicity": "100.0% Strict Monotonic",
  "outOfOrder": "0 packets (0.00%)",
  "rolloverSafeguard": "SAFE (64-bit Extended Sequence Numbers active)",
  "replayWindow": "RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted"
}
}
```

### OVERALL TECHNICAL VERDICT

Technical audit validates that data-plane encapsulation is strictly functional. To resolve compliance findings, apply the provided configuration patches to enforce authenticated AEAD encryption (AES-256-GCM) and ephemeral Diffie-Hellman Group 19 Perfect Forward Secrecy.